

Authors' Response to Reviews of

Neuromorphic Computing Based on Parametrically-Driven Oscillators and Frequency Combs

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RC: Reviewers' Comment, AR: Authors' Response, ☐ Manuscript Text

Dear Reviewers,

Thank you for the reviews. The comments led us to add several diagnostics that we agree the original submission lacked, and these have materially sharpened the mechanistic account of why the parametric resonance regime outperforms the comb regimes. All required corrections have been made to the manuscript and the supplemental material. Additions to the main text are marked in red; the supplemental material has been extended with eight new sections.

Section numbers below refer to the revised supplemental material.

1. Reviewer #1

RC: *The manuscript adopts one-step-ahead prediction of chaotic time series as the benchmark for reservoir performance. While this is a widely used and reasonable choice, it conflates several underlying computational properties, such as short-term memory and nonlinear transformation capability, into a single aggregate metric (NMSE), making it difficult to disentangle which specific computational resource is responsible for the observed differences between regimes. I would recommend that the authors additionally evaluate the information processing capacity (IPC) proposed by Dambre et al. [Sci. Rep. 2, 514 (2012)]. By decomposing reservoir capacity into contributions from different nonlinear degrees and delays, this approach would allow the authors to identify precisely which computational capabilities are enhanced or lacking in the parametric resonance regime compared to the frequency-comb regime.*

AR: We agree, and we have carried out full IPC decomposition. The results are in Sec. II of the supplemental material and are summarized in the second paragraph of page 3 of the main text.

The decomposition requires an i.i.d input, since Legendre basis is orthogonal only under that measure and driving with Mackey-Glass would violate the orthogonality the method depends on. Therefore, we used $u(n) \sim \mathcal{U}[-1, 1]$, i.i.d., with the drive built from $(u+1)/2$ and the targets taken from u directly. The basis caps total degrees at 11, delays at 25, and at most three distinct delays per target. This gives 447,161 basis functions. The estimator was validated against a 20-node linear delay line, where it returns a total of 20.175 against a measured rank of 20 (0.87% relative error), with 20,000 recovered at degree one.

	Parametric resonance	Coherent comb	Chaotic comb
Total capacity	160.32	18.0 (16.9–18.2)	5.54 (5.49–5.81)
Fraction of the bound $r = 256$	62.6%	7.0%	2.2%
Capacity per available dimension	0.626	0.070	0.022
Ratio to parametric resonance	–	$8.9\times$ lower	$29\times$ lower

Table 1: Comparison of nonlinear memory capacity across the three regimes.

The regimes differ in total capacity as well as where that capacity sits. At parametric resonance the per-degree contribution rises across the whole basis: 2.07, 3.29, 4.88, 6.92, 9.28, 12.21, 15.60, 19.35, 23.75, 28.82, 34.13. It is still rising at degree 11, the highest degree scanned. The two comb regimes behave differently from each other. The coherent comb neither deepens nor vanishes, staying flat between 1.5 and 1.8 at every degree, while the chaotic comb decays steadily and falls to within a factor of two of the false-positive expectation beyond degree three.

Both comb regimes are therefore shallower than parametric resonance, and both lose the high-degree nonlinear capability that one-step prediction of a chaotic series requires. As any finite degree cap truncates the sum, these totals are lower bounds and the measurement establishes the ordering and the accessible depth rather than a saturated total.

RC: *To better understand why the frequency-comb regime underperforms relative to naive expectations, I would suggest that the authors examine two additional diagnostic quantities: the echo-state property (ESP) and the effective dimensionality of the reservoir state. The ESP can be quantified by measuring how quickly trajectories of the reservoir, initialized from different initial conditions but driven by the same input sequence, converge onto a common trajectory; faster convergence indicates a strong echo-state property. The effective dimensionality, on the other hand, characterizes how many independent degrees of freedom are actually being used within the 2048-dimensional reservoir state and can be measured by principal component analysis. In general, good computational performance requires both a strong echo-state property and sufficiently high dimensionality, but these two quantities are known to trade off against each other, so an appropriate balance between them is essential. I expect the frequency-comb regime to exhibit higher effective dimensionality than the parametric resonance regime. However, given its inferior computational performance, I would hypothesize that this comes at the cost of a degraded echo-state property. I recommend that the authors directly measure both quantities across the sub-threshold, parametric resonance, and comb regimes, as this would provide a clear mechanistic account of the observed performance differences.*

AR: This hypothesis is what we find. Both quantities are now measured across all four regimes in Sec III of the supplemental material.

Effective dimensionality is quantified by the participation ratio $D_{\text{eff}} = (\sum \lambda)^2 / \sum \lambda^2$ over the covariance eigenvalues of the standardized feature matrix. This requires no arbitrary variance cutoff and satisfies $1 \leq D_{\text{eff}} \leq r$. The echo state property is measured by driving 12 replicas from different initial state conditions with an identical input sequence and fitting $\ln d(n) = \ln d(0) - \lambda n$ to the mean pairwise separation. The distance accounts for the Z_2 symmetry $(\psi_1, \psi_2) \rightarrow (\psi_1, -\psi_2)$ in Eqs. (1) and (2) in the main text. Without this, parametric resonance could appear to violate the echo-state property because replicas may settle on opposite branches. These branches are equivalent and cannot be distinguished by the Z_2 -invariant readout.

Table 2: Effective dimensionality, variance concentration, active features, and convergence across the four dynamical regimes.

	Sub-threshold	Parametric resonance	Coherent comb	Chaotic comb
D_{eff}	1.915	2.770	7.147	7.403
Components for 95% / 99% variance	2 / 2	8 / 11	30 / 89	54 / 179
Leading component, variance fraction	0.607	0.565	0.310	0.307
Active features (of 2048)	1024	2048	2048	2048
Convergence rate λ (per symbol)	4.99	4.03	2.5×10^{-4}	2.1×10^{-4}
Exponential fit R^2	1.000	0.974	0.011	0.004

Effective dimensionality rises monotonically into the comb regimes, $1.915 \rightarrow 2.770 \rightarrow 7.147 \rightarrow 7.403$. On the other hand, the echo-state property collapses. Parametric resonance converges rapidly, with a rate of 4.03 per symbol and an exponential fit of $R^2 = 0.974$. Neither comb regime shows clear convergence. The fitted rates of 2.5×10^{-4} and 2.1×10^{-4} per symbol have very low R^2 values of 0.011 and 0.004, respectively. Therefore, the exponential model does not describe the comb regimes well, and these rates should not be interpreted as slow convergence.

Below threshold, $\psi_1 \equiv -$, so the 1024 mode-2 columns are constant and dropped. This leaves 1024 active features. The low D_{eff} is not a consequence of having half as many active features. The constant mode-2 columns carry no variance and therefore do not contribute to the participation ratio. Instead, the low D_{eff} reflects the low dimensionality of the surviving mode-1 features: the first two components account for 60.7% and 39.3% of the variance, respectively, capturing 99.94% in total and providing $D_{\text{eff}} = 1.91$.

Between parametric resonance and the coherent comb, D_{eff} increases by $2.5\times$, while the number of components needed to capture 99% of the variance increases by $8\times$. This difference shows that much of the additional dimensionality in the comb lies in weakly excited directions. These directions contribute little to the participation ratio and are strongly suppressed by the readout regularization. Thus, the combs provide richer spectra and higher dimensionality, but with poorer state convergence, which helps explain their lower prediction performance.

RC: *In Fig. 2(a) (and also in the caption of Fig. 1), the modulation depth is stated as $F = 0$. However, according to Eq. (3), $f(n) = F_{\text{avg}} + \Delta F \times (s(n) - 0.5)$, setting $\Delta F = 0$ renders the drive amplitude $f(n)$ completely independent of the input symbol $s(n)$. Under this condition, no information about the input time series would be transmitted into the reservoir via the drive amplitude, which would make the subsequent prediction task meaningless. Since the manuscript nonetheless reports successful prediction, I suspect this is a typographical error.*

AR: The reviewer is correct that the label was wrong. We thank them for pointing this out. It was a mislabelling, not a typographical error. The input was always included in the drive.

Throughout the study, we use the input-proportional encoding

$$f(n) = F_{\text{avg}}(s(n) + 0.5), \quad (1)$$

so the drive ranges from $0.5F_{\text{avg}}$ to $1.5F_{\text{avg}}$. This is equivalent to Eq. (3) with the modulation depth set to $\Delta F = F_{\text{avg}}$:

$$F_{\text{avg}}(s(n) + 0.5) = F_{\text{avg}} + F_{\text{avg}}(s(n) - 0.5). \quad (2)$$

Thus, reporting $\Delta F = 0$ was inconsistent with the definition of ΔF as the modulation depth.

We have corrected this throughout the manuscript. All occurrences of “ $\Delta F = 0$ ” have been changed to “ $\Delta F = F_{\text{avg}}$ ”. This includes the caption of Fig. 1, the three panels of Fig. 2, and the parameter notes for Figs. 3–5. The methodology now also states that $\Delta F = F_{\text{avg}}$ is the default encoding unless otherwise specified.

This correction also required us to revise the discussion of Fig. 2(a). Since $\Delta F = F_{\text{avg}}$ varies with F_{avg} and reaches 50 along the horizontal axis, panel (a) does not represent the smallest modulation depth. Instead, its modulation depth increases with F_{avg} . We have therefore removed the statement that ΔF increases from zero and clarified that using a fixed ΔF removes this dependence on F_{avg} , producing a more uniform low-NMSE region.

RC: *In constructing the reservoir feature vector, the authors include the discrete Fourier transform (DFT) of the sampled ψ_1 and ψ_2 trajectories, in addition to their time-domain trajectories. It is not entirely clear from the text whether this DFT is computed over the short trajectory sampled within a single symbol period of length T_{symbol} , or over the full time series spanning multiple symbol periods. I would ask the authors to explicitly clarify the exact time window over which this DFT is computed.*

AR: The transform is computed over a **single symbol period**, not the full record. We have clarified this in the methodology and in Sec. IV of the supplemental material.

For each symbol n and mode $j = 1, 2$, the transform uses the same $N_v = 512$ virtual-node samples as the time-domain features,

$$\tau_k = \frac{kT_{\text{symbol}}}{N_v}, \quad (3)$$

with $T_{\text{symbol}} = 5.0$ in normalized time. The frequency-bin spacing is therefore $1/T_{\text{symbol}} = 0.2\tau^{-1}$. Thus, the transform does not capture timescales longer than one symbol; any memory across symbols is carried by the state ψ .

The transform is applied to the **complex modal amplitudes** ψ_1 and ψ_2 , not to their sampled powers. The spectral features are

$$|\mathcal{F}[\psi_j]|, \quad (4)$$

with the magnitude taken after the transform. This preserves intra-symbol phase information that is lost in $|\psi|^2$. Since ψ_j is complex, its spectrum is two-sided and asymmetric, so the positive and negative frequency bins can contain different information. With $N_{\text{FFT}} = N_v$, the full two-sided spectrum is retained. If the transform were instead applied to the sampled power, the signal would be real and its spectrum would be conjugate-symmetric, making the spectral features largely redundant with the time-domain features.

RC: *The governing equations (1) and (2) are presented without an explicit derivation. Given that these equations form the theoretical foundation of the entire study, I would recommend that the authors include a detailed, self-contained derivation in the supplemental material.*

AR: A complete, self-contained derivation of Eqs. (1)–(2) in the main text has been added to Sec. I of the supplementary material.

RC: *The manuscript does not explicitly state whether, during the testing phase, the predicted output is fed back into the drive amplitude encoding to autonomously generate the subsequent time steps in a closed-loop (free-running) fashion, or whether the true time series is used to drive the reservoir at every time step even during testing (open-loop prediction). This distinction is important for interpreting the reported NMSE values: closed-loop generation would be expected to accumulate error over time, whereas open-loop evaluation measures only the local, single-step predictive accuracy of the readout at each time point. I would ask the authors to explicitly clarify which protocol was used.*

AR: The prediction is performed in an **open-loop (teacher-forced)** setting. The true time series is used to drive the reservoir during both training and testing, and the predicted output is never fed back into the drive. Thus, the reported NMSE measures single-step prediction error and does not include error accumulation from free-running prediction. We have clarified this in the methodology.

2. Reviewer #2

RC: *7. Redundancy in feature extraction: The feature vector combines time-domain samples (1024 dimensions) and Fourier magnitudes (1024 dimensions) from the same symbol period trajectories. Unless the sampling rate violates Nyquist, the spectral features introduce little independent information. The authors should demonstrate their added value via ablation studies or rank analysis; otherwise, the high dimensionality (2048) risks overfitting despite ridge regularization.*

AR:

RC: *8. No quantitative criterion for distinguishing coherent vs. chaotic combs: The boundary between these regimes is not defined. A metric (e.g., phase diffusion coefficient, spectral line width, or Lyapunov exponent) must be specified, and the exact parameter values for the chaotic-comb examples in Fig. 1(k-o) must be provided in the caption.*

AR:

RC: *9. Data rate not rescaled to the intrinsic timescales of benchmark systems: The Mackey-Glass, Rössler, and Lorenz systems have widely different characteristic times (nearly an order of magnitude), yet all are fed at the same rate of 2424 samples/s. This leads to vastly different effective memory capacities across tasks. The authors should either rescale the symbol duration for each system or explicitly justify the single rate and discuss its impact.*

AR:

RC: *10. Initiation protocol in the bistable region not quantified: The authors note that performance in the bistable region depends on initialization, but they do not report how often the 50-period constant-amplitude drive actually selects the parametric branch. A sensitivity analysis to initial conditions and a more robust startup method (e.g., chirped drive) should be considered.*

AR:

RC: *Insufficient details on regularization and numerical integration: The ridge penalty $\lambda = 10^{-3}$ is fixed without cross-validation, despite the feature dimension (2048) greatly exceeding the training sample size (800). The choice of λ critically affects generalization. Also, the adaptive Runge-Kutta integrator is not verified for stiff regimes (e.g., chaotic combs); the error tolerance and steps per symbol period should be reported.*

AR: